
Systematic Trading in Bitcoin Prediction Markets

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Abstract

This report summarises an independent research project on Polymarket’s short-duration Bitcoin Up/Down contracts. I reconstructed the contracts’ external reference price from multi-venue cryptocurrency data, tested statistical and machine-learning strategies, and deployed selected approaches in a live event-driven trading system. Live testing showed that execution latency could remove an apparent backtested edge, shifting the research focus from predictive accuracy towards latency-aware expected value.

1 Motivation and Research Process

Prediction markets provide a compact setting in which probability modelling, market microstructure and execution interact directly. I initially set out to understand whether Polymarket’s Bitcoin contracts could be modelled systematically, with the longer-term aim of extending the framework to other contract horizons and underlying assets.

Rather than begin with a preferred model, I followed an end-to-end research process:

market mechanics → data → strategy research → backtesting → live execution → iteration.

The central lesson was that a signal is only useful if it survives data limitations, transaction costs and executable latency.

2 Resolution Modelling and Data

The contracts resolve against a Chainlink reference price constructed from multiple undisclosed data sources, rather than a print from one exchange. I therefore built a resolution-window collector covering 17 BTC quote pairs across 12 exchanges. It sampled top-of-book data every 25 ms, stored top-10 book snapshots when venues changed, and normalised USD, USDT and USDC quotes onto a common dollar basis.

The proxy was evaluated against completed Polymarket outcomes. On the 15-minute dataset, 227 boundaries produced 206 matched contracts and 412 endpoint observations. The best five-venue combination achieved an MAE of \$2.34 and RMSE of \$3.04. On a later five-minute sample of 282 resolutions, a five-venue proxy achieved an MAE of \$1.53 and RMSE of \$2.31.

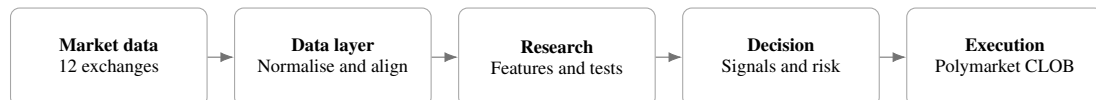


Figure 1: Research and live-execution pipeline.

3 Strategy Research

I compared rule-based strategies—including scalping, market making and statistical arbitrage—with Q-learning and LSTM-based approaches. Strategies were assessed on signal stability, downside

exposure, transaction costs and whether the decision could be executed under live market conditions. Only the more viable, lower-risk candidates were progressed to deployment.

The system detected directional moves over rolling windows, applied contract-time, price and adverse-move filters, and required a fresh threshold move after a blocked or consumed signal to avoid trading stale events.

4 Live System and Execution Research

I implemented the selected strategies as an asynchronous, event-driven Python system. Polymarket books were maintained from its market WebSocket after a REST snapshot, with reconnection and snapshot fallbacks. Orders were submitted as fill-and-kill limits through the CLOB.

The implementation included:

- cached signer and token-risk state, persistent HTTP sessions and warmed connections;
- separate worker pools for general CLOB calls, entries and urgent exits;
- controls for duplicate orders, in-flight entries, stops and failed-sell retries;
- delayed on-chain balance reconciliation to detect fills after a timeout; and
- structured logging of book age, signing and post times, signal-to-fill latency, slippage, fees and net P&L.

Analysis of 2,394 historical order attempts quantified the main execution constraint. The fill rate was 50.7%, median HTTP post duration was 1,799 ms, and the median upper bound on signal-to-fill time for filled orders was 1,825 ms. Filled orders executed a median \$0.02 above the signal-time ask. For failed orders, the median maximum ask increase while the request was pending was \$0.08, compared with \$0.03 for filled orders.

Table 1: Selected measured results from the research and live-system logs.

Metric	Result
Resolution collector	17 pairs across 12 exchanges
15-minute proxy validation	412 endpoints; MAE \$2.34
5-minute proxy validation	282 resolutions; MAE \$1.53
Historical order attempts	2,394
Observed fill rate	50.7%
Median HTTP post duration	1,799 ms
Median filled signal-to-fill upper bound	1,825 ms

These observations changed the trading objective. A signal should be evaluated against the likely price after execution delay, not only the price visible when the signal is generated.

5 Ongoing Research

Current work focuses on strategies that retain positive expected value under observed execution conditions:

- **Latency-aware backtesting:** sampling empirical order delays and replaying the intervening order-book path rather than assuming immediate fills.
- **Signal-decay modelling:** estimating how quickly the edge following an external exchange move disappears and rejecting signals whose expected half-life is shorter than the execution path.
- **Execution improvements:** testing pre-signed orders, persistent connections, evaluating geographically closer VPS deployment to reduce order-path latency and maker execution where fill probability justifies the lower cost.
- **Robust strategy selection:** comparing entry filters, stop rules, position caps and drawdown behaviour after fees using replay and counterfactual analysis.
- **Broader markets:** extending the resolution proxy and strategy framework across five- and 15-minute contracts, longer-duration markets and other underlying assets as reliable data becomes available.